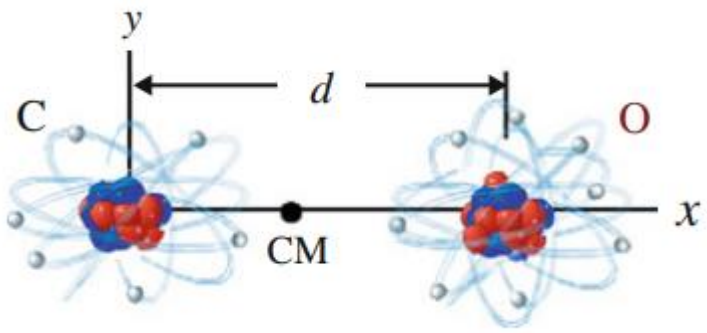
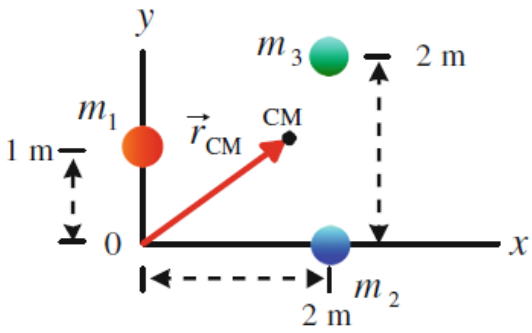
Exercises on the motion of systems of particles: Center of mass and Moment of inertia

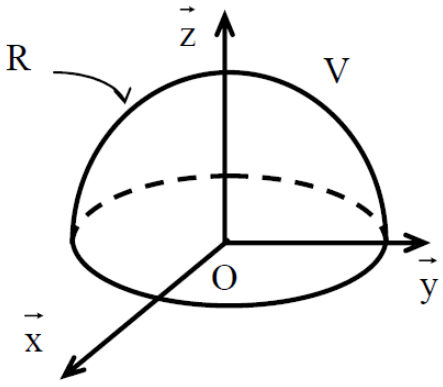
1. An oxygen atom ($^{16}_{8}O$) has a mass m_O = 16u, and a carbon atom $^{12}_{6}C$ has a mass m_C = 12u. The average distance between their nuclei in the CO molecule is d = 0.113 nm, see Figure bellow. How far from the oxygen nucleus is the center of mass of the molecule?

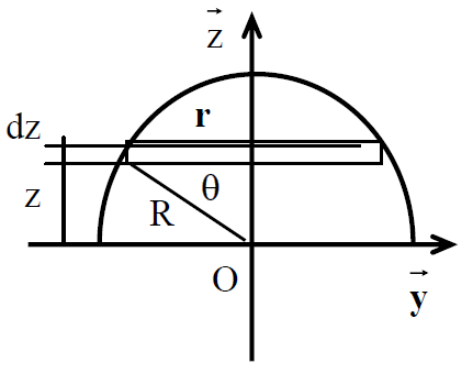


2. A system of three particles of masses m_1 = 2.5kg; m_2 = 1kg; and m3 = 1.5kg are spread out in two dimensions and located as shown in the figure bellow. Find the center of mass of the system.



3. Determine the position of the center of mass of a full homogeneous half sphere of radius R





4. A mass $m_1 = 2kg$ is connected to a mass $m_2 = 3kg$ by a massless rod. The location of m_1 and m_2 are given by the position vectors $\vec{r}_1 = 4\vec{i} + 5\vec{j}$ and $\vec{r}_2 = 2\vec{i} + 3\vec{j}$ respectively. Find the coordinates of the center of mass.

5. Given A horizontal rod has a mass M and length L. Find the location of its center of mass fromits left end:

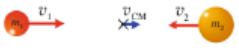
(a) if the rod has uniform mass per unit length λ and
(b) if the rod has a mass per unit length that increases linearly from its left end according to the relation $\lambda = \alpha x$ where α is a constant.

6.

A ball of mass $m_1 = 2kg$ traveling with velocity $\vec{v}_1 = 15\vec{i}(m/s)$ collides head on and elastically with a second ball of mass $m_1 = 3kg$ traveling with velocity $\vec{v}_2 = -4\vec{i}$ m/s, see Figure bellow.

(a) Find the velocities of the two balls after the collision.

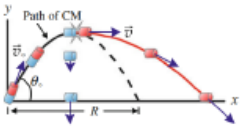
(b) Find the velocity of the center of mass before and after the collision.



7.

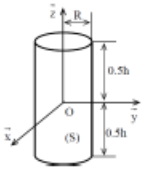
A projectile is fired from the ground with an initial speed v_o of 40 m/s at an angle θ_0 of $15^o m/s$ above the horizontal direction. At the maximum height, the projectile explodes into two fragments of equal mass, see Figure bellow. One fragment stops momentarily and falls vertically, while the second one flies initially in a horizontal direction.

How far from the ground do the center of mass and the second fragment land? Take $g = 10m/s^2$.



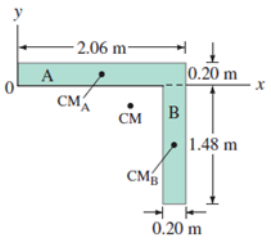
8.

Determine the inertia matrix in O, with respect to an orthonormal coordinate system (O, x,y,z), of a homogeneous cylinder (S), of center O, and of radius R



9. Prove that the momentum of a system of N particles is the same as that of a single particle whose mass is that of the whole system, moving with the velocity of the center of mass. Show again that it follows that the velkocity of the CM is constant if the system's momentum is conserved.
10. Given A homogeneous horizontal rod has a mass M and length L. Find moment of inertia about an axis passing through its center of mass
11. The moment of inertia of circular loop of radius R, at a distance of R/2 aound a rotating axiks parallel to horizontal diameter of the loop is
    a) $\frac{1}{2}MR^2$
    b) $\frac{3}{4}MR^2$
    c) MR^2
    d) $2MR^2$

12. Determine the CM of the uniform thin L-shaped construction brace shown in figure bellow

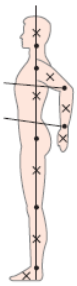


This L-shaped object has uniform thickness d (not shown)

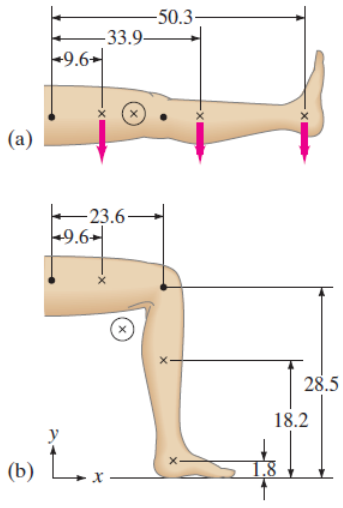
13. we consider the human body. Table 7–1 indicates the CM and hinge points (joints) for the different components of a "representative" person. Of course, there are wide variations among people, so these data represent only a very rough average. The numbers represent a *percentage* of the total height, which is regarded as 100 units; similarly, the total mass is 100 units. For example, if a person is 1.70 m tall, his or her shoulder joint would be (1.70 m)(81.2/100) = 1.38 m above the floor.

TABLE 7–1 Center of Mass of Parts of Typical Human Body, given as %
(full height and mass = 100 units)

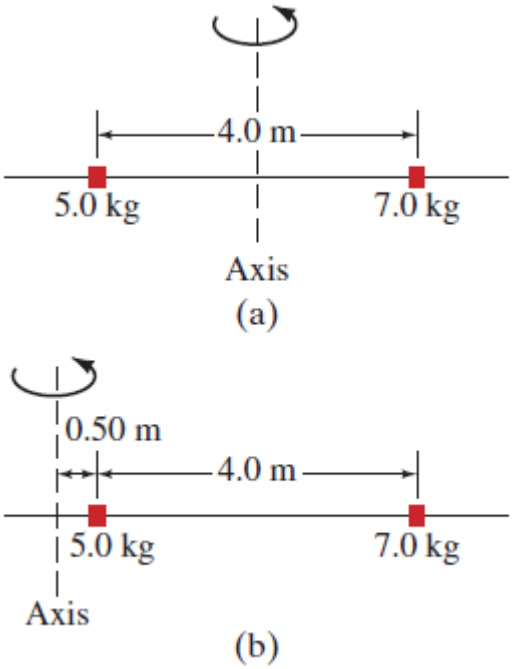
Distance of Hinge Points from Floor (%)	Hinge Points (•) (Joints)	Center of Mass (×) (% Height Above Floor)		Percent Mass
		Head	93.5%	6.9%
		Trunk and neck	71.1%	46.1%
91.2%	Base of skull on spine	Upper arms	71.7%	6.6%
81.2%	Shoulder joint	Lower arms	55.3%	4.2%
	elbow 62.2%‡	Hands	43.1%	1.7%
	wrist 46.2%‡	Upper legs (thighs)	42.5%	21.5%
52.1%	Hip joint	Lower legs	18.2%	9.6%
28.5%	Knee joint	Feet	1.8%	3.4%
4.0%	Ankle joint	Body CM =	58.0%	100.0%



Determine the position of the CM of a whole leg
(*a*) when stretched out, and
(*b*) when bent at 90°. See Figure bellow. Assume the person is 1.70 m tall.



14. Two small "weights," of mass 5.0 kg and 7.0 kg, are mounted 4.0 m apart on a light rod (whose mass can be ignored), as shown in Figure bellow. Calculate the moment of inertia of the system (*a*) when rotated about an axis halfway between the weights, Fig. a, and (*b*) when rotated about an axis 0.50 m to the left of the 5.0-kg mass (Fig. b).



15

The four masses in Fig. 11-8 are held rigid by the very light circular frame shown. Find the moment of inertia and radius of gyration of the system for an axis through the center of the circle and perpendicular to the page.

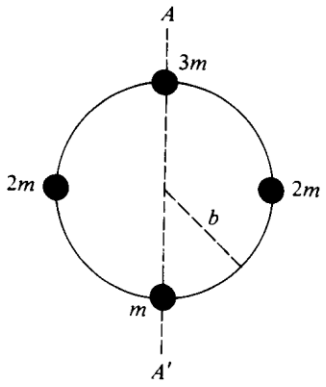


Fig. 11-8

16.

Using integration, compute the moment of inertia of a uniform thin rod of length L about an axis perpendicular to the rod at a point $L/4$ from one of its ends. Show how the same result could be obtained by use of the parallel-axis theorem.

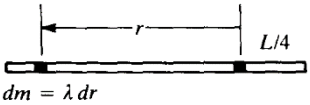


Fig. 11-12